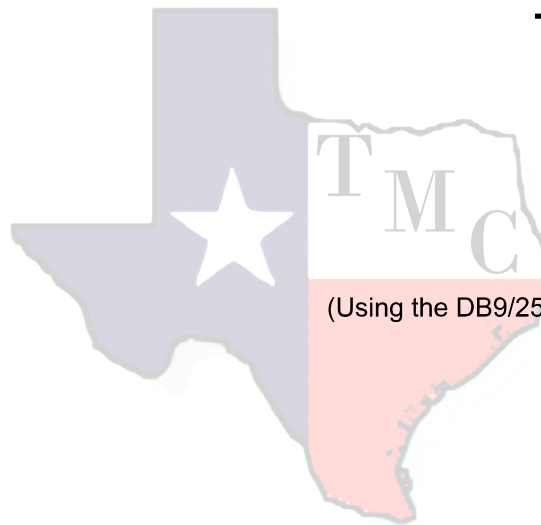




Texas MicroCircuits



Torch Height Controller Quick Start

Models THC301d & THC301b

Bench Testing & Installation

(Using the DB9/25 Test Plugs, and Clip Leads for Testing)

Rev 1.02

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Revisions

Version 1.00 First Release 07/11

Version 1.01 Fixed typos and grammar 08/11

Version 1.02 Added calibration during testing and removing resistor jumper

Firmware Versions

Almost all of the THC301 models were at firmware 1.0 for both the logic board and display board before 5/2011. When you powered the THC301 up, it displayed the Set Volts only. With the introduction of newer versions of both the logic board and display board firmware, when powering up the THC, it would first show the display version number (either 255 or 512) for a second or so, then display the logic board version (12x or 13x at the present) for a second or so and then settle on displaying the Set Volts. (The “x” above is the last digit and is disregarded. 12x refers to version 1.2 and 13x refers to version 1.3)

Logic Board

Version 1.0 and 1.1 were introduced before 9/2010

Version 1.2 5/15/2011 This fixed a problem with the anti-dive algorithm, especially while cutting very thin material at high speeds.

Version 1.3 7/15/2011 This version fixed a delay of the THC becoming active quick enough after piercing the material

Display Board

Version (nothing) All THCs before 1st quarter of 2011

Version 255 This is not really a version change but making the display show whether it will display 0 to 255 volts. All THCs that are NOT hi-def machines use this version.

Version 512 On the high end Hypertherm and Thermal Dynamics machines, they can have tip voltages in excess of 255 (up around 450 or more). This version allows the display to show 0 to 512 volts. WE install this version when so requested by the user because he will use it on a hi-def machine.

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If You Are Not Planning to Read This Quick Start Manual...

I strongly urge you to reconsider...PERIOD! This is especially true if this is the first time you are installing and using a Torch Height Controller (THC) on a CNC system. This Quick Start manual will cover bench testing to give you confidence that each unit is operational, and installation of hardware and software. Much can go wrong if improperly installed and understanding its proper operation will save you a lot of headaches down the road. If any tests in this Quick Start manual fail, please refer to the more detailed User Guide for help in isolating and repairing the problem.

Read This Before You Begin... (The proverbial “fine print”)

Any computer controlled machine tool is potentially dangerous. You’ve already accepted responsibility for the operation of the CNC machine you expect to control with the THC. The THC is a device that was designed to control the Z axis height while running the Mach © software and cutting metal using a plasma cutter.

The Mach © software runs a myriad of different types of machines worldwide. This manual is here to give you guidance on safety issues and proper operation of the THC and associated options. Since we do not know the details of your machine, environment, or operator skill level, we cannot take responsibility for any damage or injury caused by its use. It is your responsibility that you understand your machine, its capabilities, and any safety issues it generates.

If you are in any doubt, be sure to seek guidance from a professionally qualified expert rather than risk injury to yourself, others, or damage to your equipment.

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Texas MicroCircuits is not responsible for any damage to equipment or tools as the result of usage of the THC devices.

Quick Start Overview and Assumptions

Overview

This document is a Quick Start manual for the Torch Height Controller (THC) model 301 both “d” and “b” versions. The device is designed by Sound Logic and manufactured by Texas MicroCircuits in the USA. The only real difference in the THC301 models is the THC301d model comes mounted in its own enclosure whereas the THC301b does not. The engraved panels for both are identical in function but are different sizes. Other than that, both models are identical. We are updating some user functions soon like converting to 1 adjusting multi-turn knob (instead of separate Coarse and Fine knobs) and a textured panel. The old layout with 2 knobs will still be available upon request. See photos below.

Note: **Black** text is information, Blue text is instructions, and Red text is important!

Block Diagram and Overview of the CNC Plasma System with THC

The block diagram showing the CNC plasma system with THC is a high level diagram showing the major components:

- PC running Mach3 with 2 printer (parallel) ports
- Torch Height Controller (either THC301d or THC301b)
- Sensor Board with Hall Effect Current Sensor (HECS) if needed
- Drive electronics including, breakout board, optional relays and status board, stepper/servo drivers, power supply, and motors
- Plasma machine
- Z axis “floating head” touch-off mechanism or similar function.

Below is a “big picture” block diagram of the CNC and THC plasma cutting system.

Block Diagram of Plasma System w/Torch Height Controller

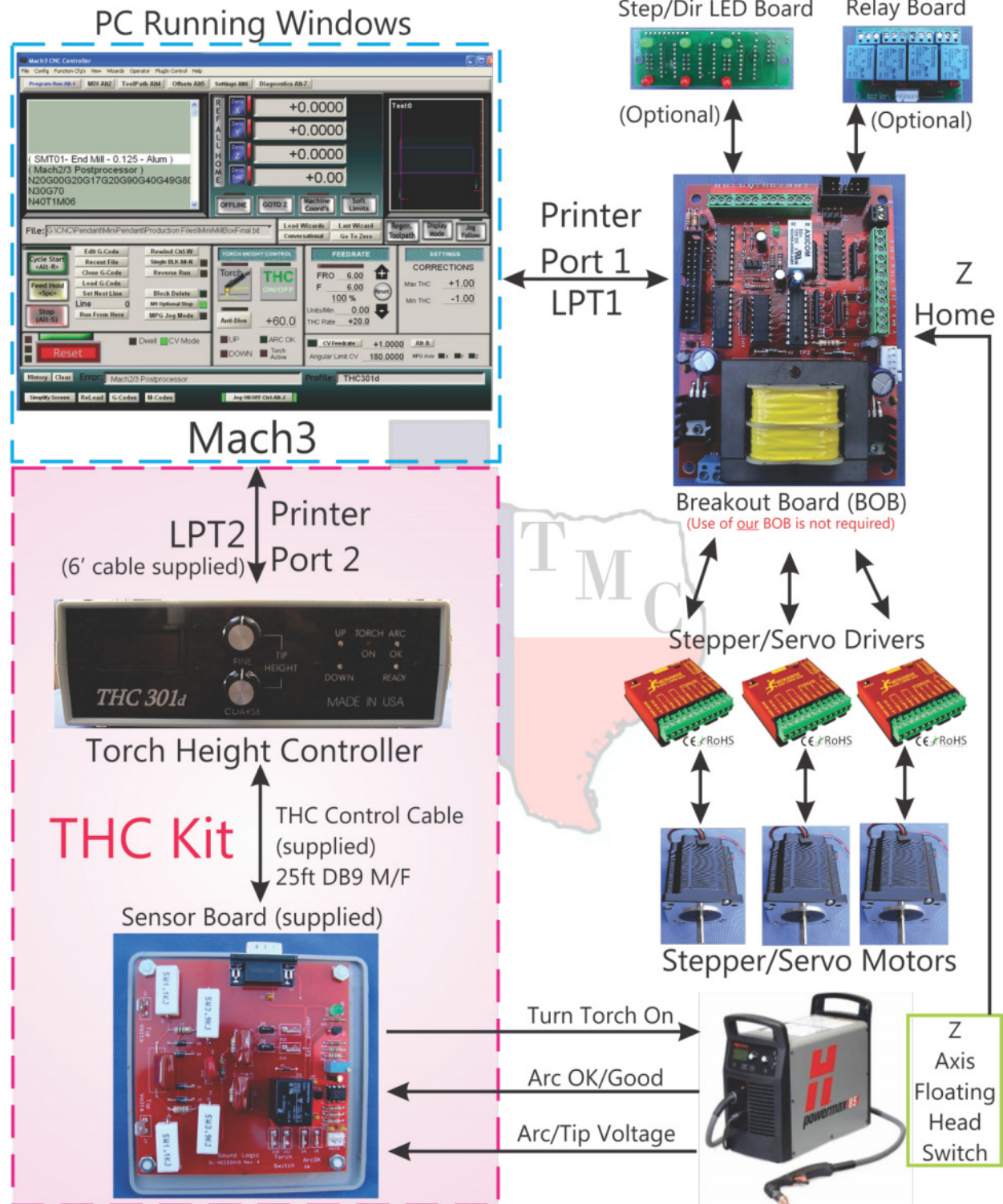


Figure 1 – Block Diagram of the CNC Plasma System with THC

Assumptions

Because this is a Quick Start guide, there are certain assumptions we have to make before you start testing and using your THC. If **ANY** of these assumptions are incorrect, it is imperative that you abandon use of this guide and refer to the User Guide for more detail on your testing and installation.

Assumption:

- You are using a THC301 (b or d), not a THC300
- You have all of the components of the THC Test Set (see Fig 1 photo below)
- During calibration, we will be using the Sensor Board internal divider (Raw Tip Volts), not a divider ratio from the Plasma cutter.
- Your PC parallel (printer) ports have been installed correctly and tested independently.
- We are **ONLY** testing the THC part of the system (Pink area of “Big Picture” Block diagram – Fig. 2) We are NOT testing the PC, parallel ports, breakout board, motors drivers, or motors. That is not within the scope of this document.
- Your THC has not been calibrated for a “high def” machine. (You would have requested this).

THC Test Set

The THC Test Set will allow **complete testing** of the THC301 and all of its units including THC301, Sensor Board, as well as the 25ft DB9 cable, and 6ft DB25 parallel port cable.

The THC Test Set consists of:

- 5 colored test clip leads (Black, Red, Green, Yellow, and White)
- 1 female to female DB25 “gender changer” (for testing the parallel port cable)
- 1 DB25 Test Plug for simulating the Mach3 connection to the THC301
- 1 DB9 Test Plug for simulating the Sensor Board connection to the THC301



Figure 2 - THC Test Set

Bench Test the THC301 in 10 Minutes or Less

This is the section you've been looking for. IF you read the entire section FIRST, then it should only take 10 minutes or less to test the THC.

In this section, using the components of the THC Test Set, I'll be able to show you how to completely test and calibrate a THC301 system from the parallel port cable that connects to your PC to the THC to the serial cable that connects to the Sensor Board and finally the Sensor board itself. We'll do ALL of this with no connection to the PC or Mach3 and no connection to your plasma cutter. Below are both a photo of a THC301b and a THC301d. Bench testing is the same for both but we'll use the THC301b for our examples because it is easier to photograph all connections for the bench testing procedure.



Figure 3 - THC301b and THC301d

Upgrading Microcontroller and Firmware

If you received a new microcontroller with your THC Test Set, please install it now. The latest firmware version is 1.3 and will show in the display (**13x** where x=last digit and ignored) when the THC is powered up. However, 1.3 changes only affect certain conditions and those with version 1.2 should be fine. Others with a version that does NOT show in the display (version 1.0) are still valid if you have not encountered any cutting issues. However, you can request the latest firmware upgrade and it will only cost you for shipping (US\$6 for US, US\$18.00 for all others).

- Locate the PIC microcontroller on the Logic Board (THC301d users will need to remove the top cover – 2 screws on the bottom of the case)

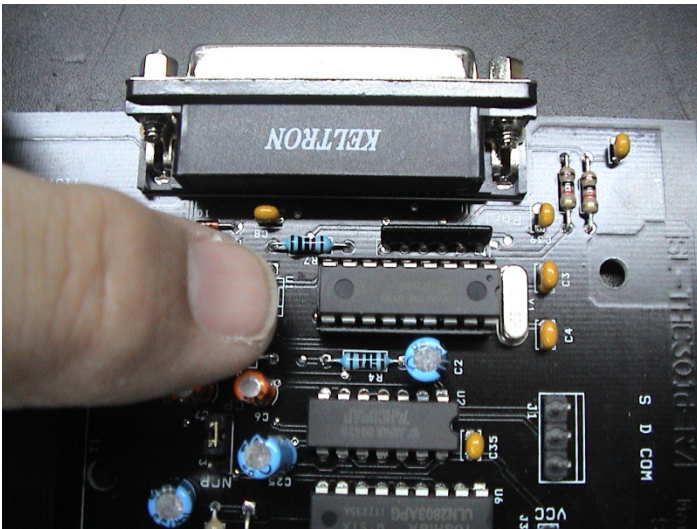


Figure 4 - Locate PIC Microcontroller

- Taking a thin flat bladed screwdriver, gently wiggle it under the end of the PIC and “walk” it on down the length of the PIC, raising it out of its socket. THC301d users will have a little more difficulty with the screwdriver hitting the sides of the enclosure, but it will work.

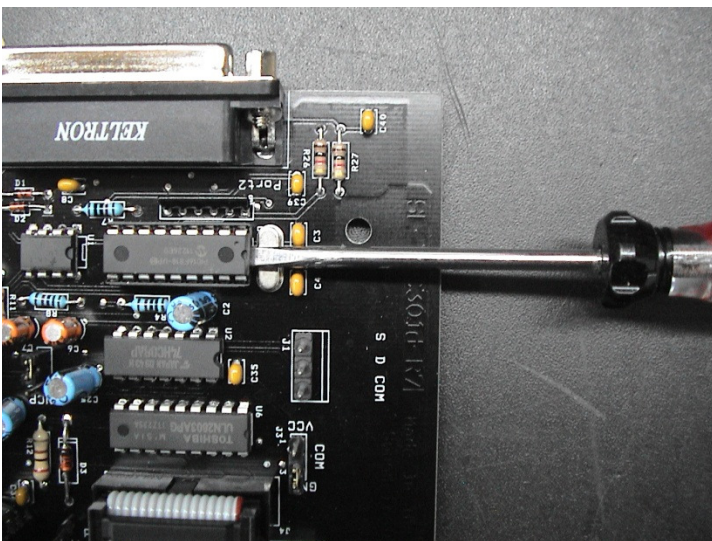


Figure 5 - Remove it with a Thin Screwdriver

- Figure 6 below shows the empty PIC socket after removal.

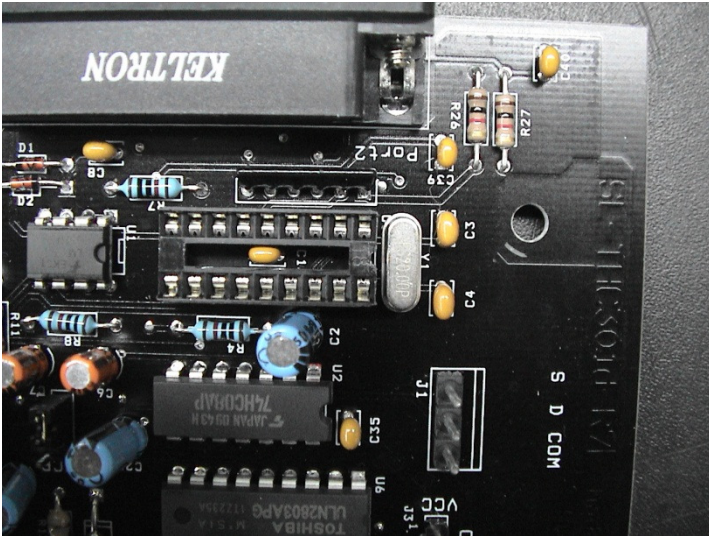


Figure 6 - Showing empty PIC Socket

- Now gently place one side of the pins into the socket, making sure they are aligned up properly. Then rock the pins on the other side into the socket until you feel both sides are lined up.

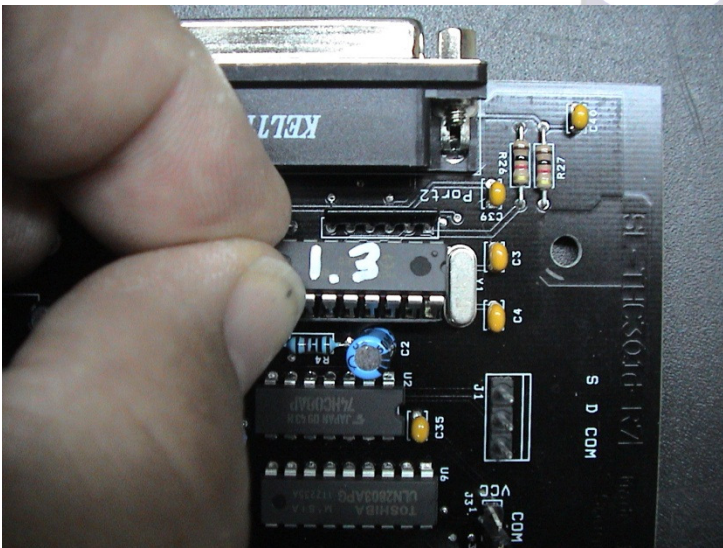


Figure 7 - Install New PIC Microcontroller

- Now press firmly the PIC controller into the socket until you feel the pins are seated. Look for any bent pins. If you find some/one, remove the chip and straighten the pin, then reinstall it.

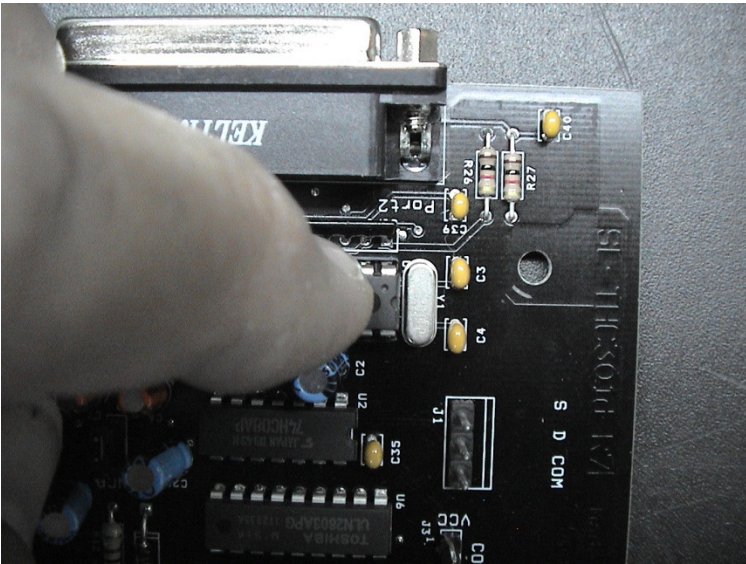


Figure 8 - Press PIC into Socket

The Display firmware has not been changed, only the later displays show the version (**255**). If your display does not show the version, there is no need to change the microcontroller on the display board.

Removing Resistor Jumper

Since you have the THC enclosure open anyway, some THC301 later models had a jumper installed on J7 between pin 1 and pin 6. This jumper was used only in some early trials for externally testing the THC and the new THC Test Set replaces the need for this jumper and it may actually impede proper testing by leaving it, so we suggest removing it. Below is a photo of its location.



Figure 9 - Resistor Jumper on J7

Just grab the black shrink wrap wire on pins 1 and 6 and pull straight up to remove. Discard jumper.

Testing the THC301 Main Unit

If you are using a THC301b, set up the unit with a power cord (but don't power it on, yet) and plug the front panel into the 2x8 connector on the logic board. If using a THC with 2 knobs, set the Coarse and Fine knobs to the center (12 o'clock). On the THC301d with a single knob, it doesn't matter on the position.

- Plug the DB9 Test Plug into the DB9 connector on the rear of the logic board.

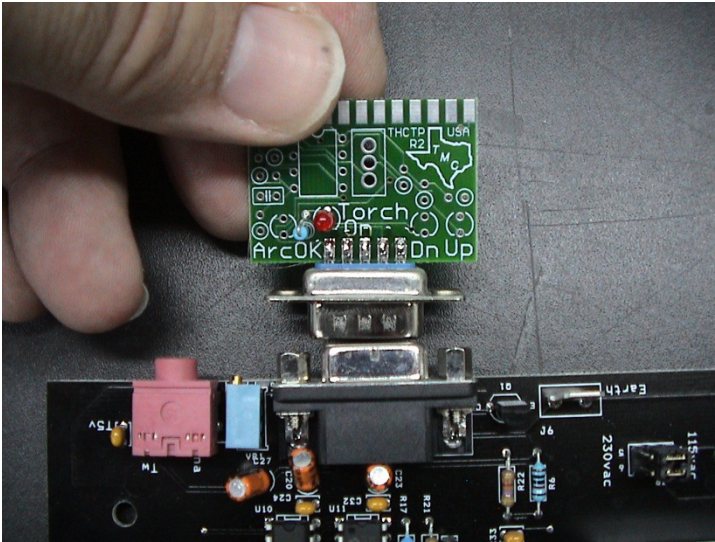


Figure 10 - Insert DB9 Test Plug

- Plug the DB25 Test Plug into the DB25 connector on the rear of the logic board.
- Make sure the toggle switch is flipped toward the logic board. (Torch Off)

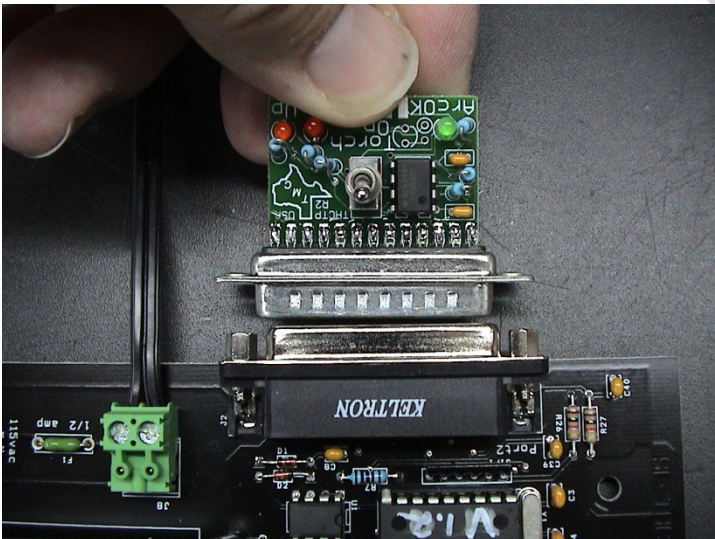


Figure 11 - Insert DB25 Test Plug (note toggle switch position...Torch Off)

- Before you apply power to the THC301, let me tell you what you are going to see as it will all happen in a few seconds and I want to prepare you for it.

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- When power is applied:
 - o The display will show **255** for a second or two. This is the firmware version of the display.



Figure 12 - Show Display Firmware Version

- o Next, it will show **13x** (the last digit can be anything from 0 to 9 and does not matter). This is the firmware version of the logic board – version **1.3**.



Figure 13 - Show Logic Board Firmware Version

- o Lastly, it will settle on some number between **0** and **255** (we will adjust this).
- The **ONLY** LEDs on should be the green power on LED on the logic board (THC301d users don't care) and the **"READY"** LED on the front panel.

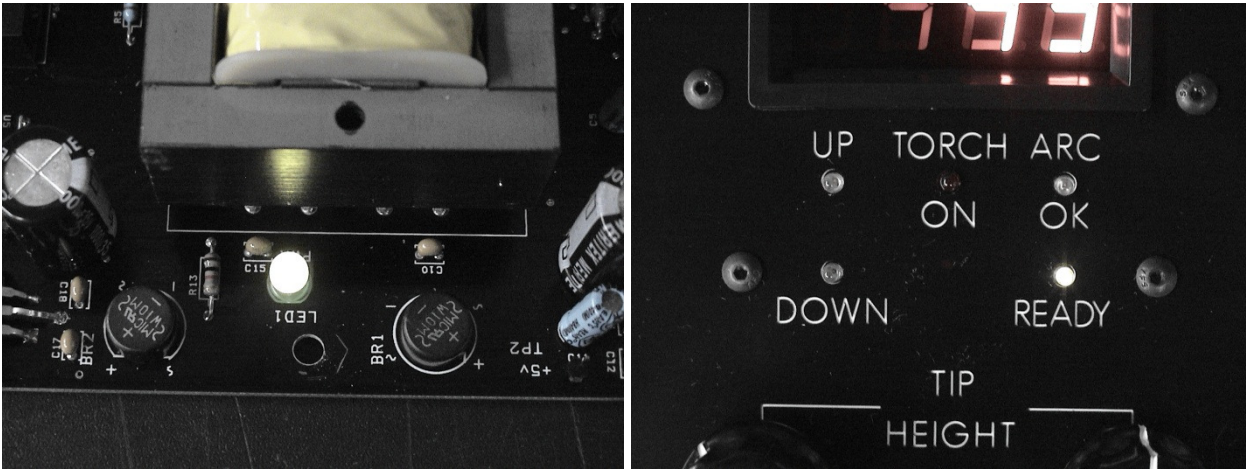


Figure 14 - Power On and Ready LED

- Turn the Single Big knob or Coarse knob on the front panel until the display shows **93**.



Figure 15 - Adjust to Set Volts of 93

- Flip the toggle switch on the DB25 Test Plug (away from the logic board).

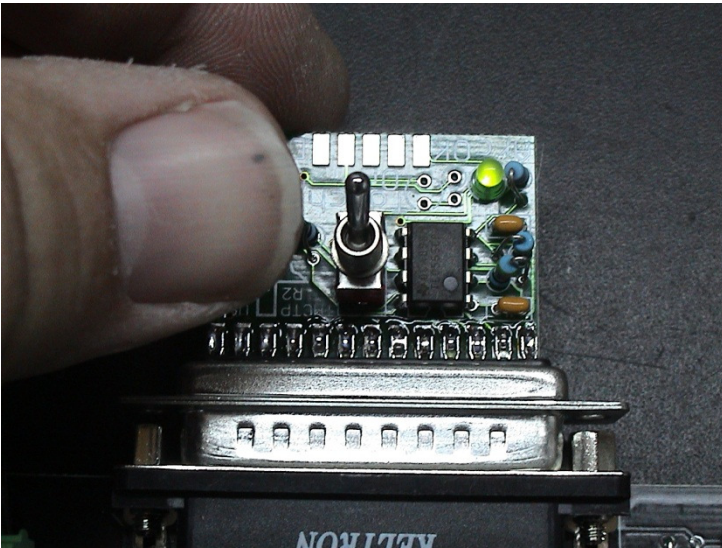


Figure 16 - "Fire" the Torch

- The "ArcOK" and "Torch On" (front panel) should turn on and the "Torch ON" (DB9 Test Plug) and the "ArcOK" (DB25 Test Plug) should turn on.

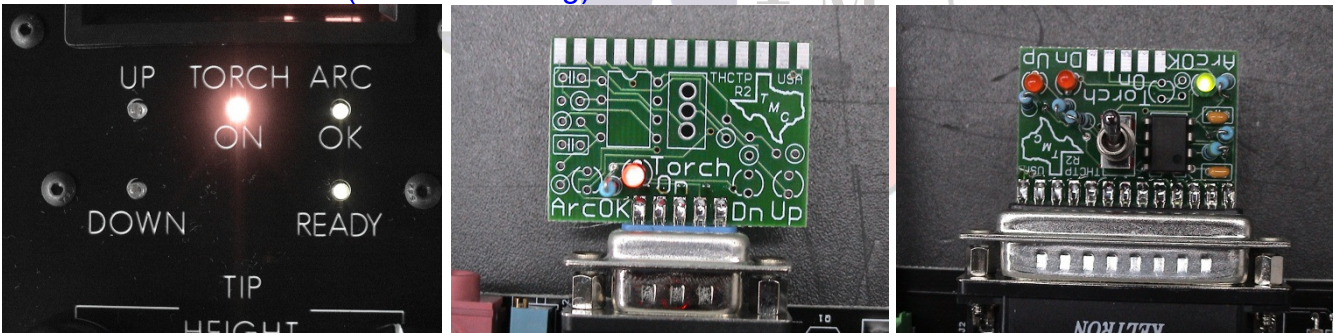


Figure 17 - Show ArcOK and TorchOn

- If your panel does not read **93** then now is a good time to calibrate The THC Tip Volts by adjusting the VR1 pot with a little flat blade screwdriver like shown in the photo below. Simply adjust the pot for a reading of **93** and you're ready to move on with testing.

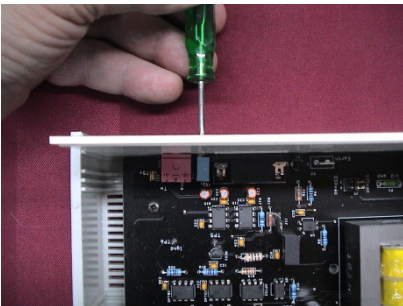


Figure 18 - Calibrating the THC Tip Volts

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- Now, carefully adjust the large Multi-turn knob or the Coarse knob clockwise until the **UP** (front panel) and **Up** (DB25 Test Plug) come on. This tells Mach to raise the Z axis.

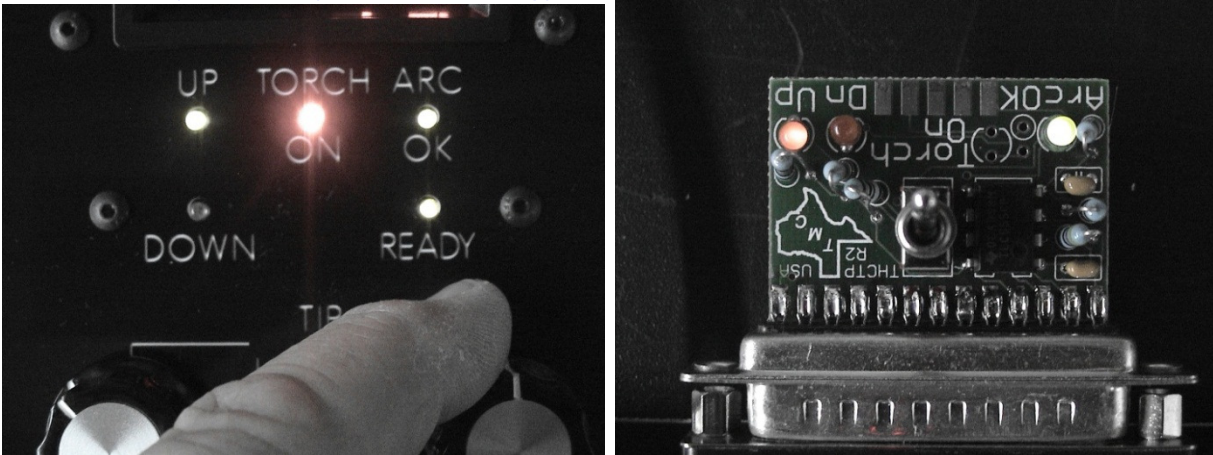


Figure 19 - Raise the Z Axis

Now carefully adjust the large Multi-turn knob or Coarse knob counterclockwise until the **DOWN** (front panel) and **Dn** (DB25 Test Plug) come on. This tells Mach to lower the Z axis.

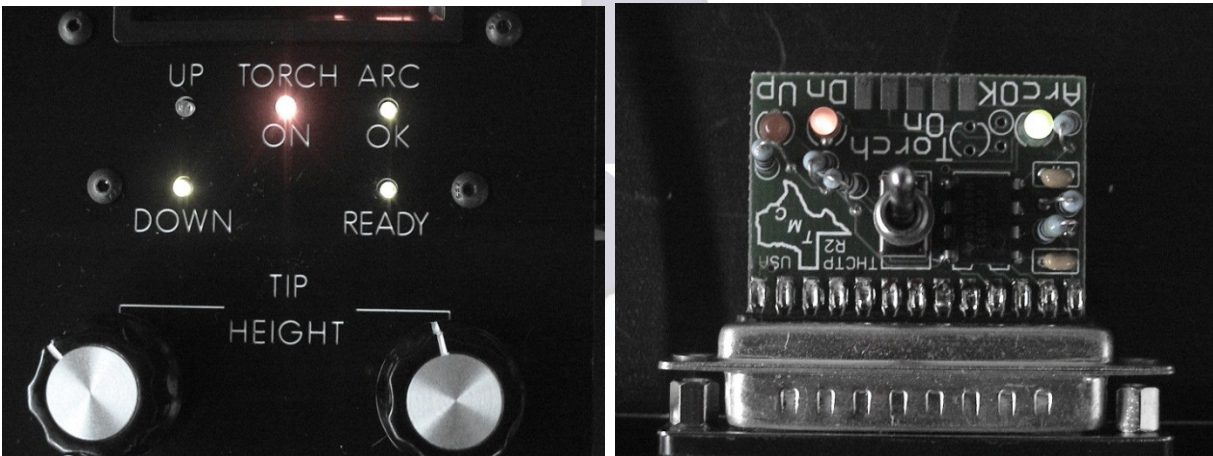


Figure 20 - Lower the Z Axis

- Finally, continue adjusting the large Multi-turn knob or the Coarse knob counterclockwise until the **DOWN** (front panel) and **Dn** (DB25 Test Plug) go out and within a 3 seconds the **"Torch On"** (front panel) and the **"Torch ON"** (DB9 Test Plug) will go out. This tells Mach that a "dive" condition exists and stops the torch and Z movement.

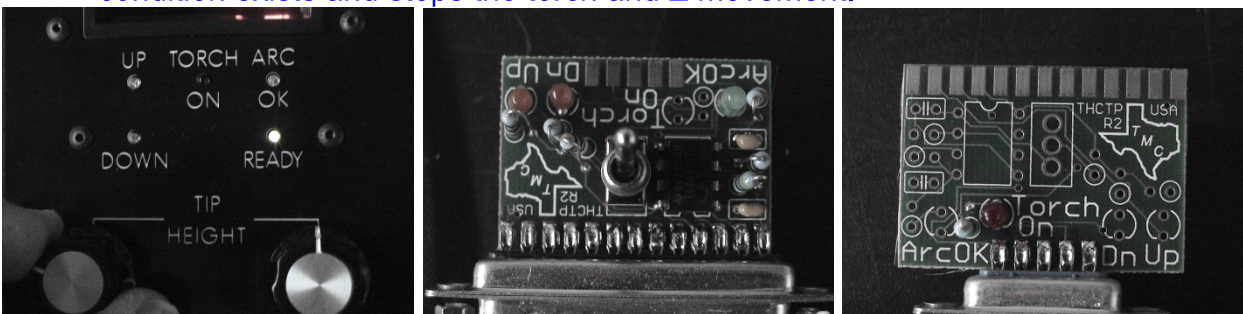


Figure 21 - Generate a "Dive" Condition

- That's it! You have now tested the main THC301 unit completely.

Testing the 6ft Parallel Port Cable

Now that your THC has been tested, just unplug the DB25 Test Plug, plug ne end of the 6ft Parallel Port cable in the DB25 connector on the logic board and plug the DB25 test Plug into the other end of the 6ft Parallel Port cable using the female-to-female gender changer and repeat the THC Bench Test. Now you have tested the parallel port cable.

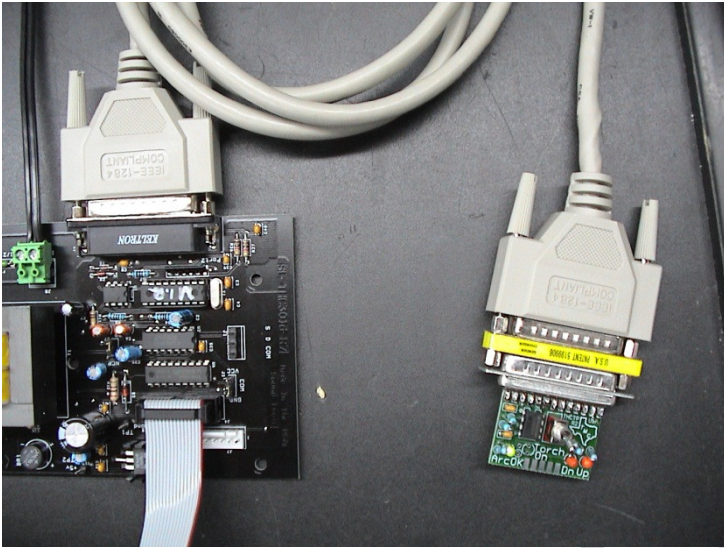


Figure 22 - Install DB25 Test Plug in 6ft Parallel Cable (and female to female gender changer)

Testing the 25ft Sensor Board Cable

Just like testing the parallel port cable above, remove the DB9 Test Plug from the logic board. Plug one end of the 25ft Sensor Board cable into the DB9 on the logic board. Plug the SB9 Test Plug into the other end of the 25ft Sensor Board cable and repeat the THC Bench Test. Now you have tested the s5ft Sensor Board cable.

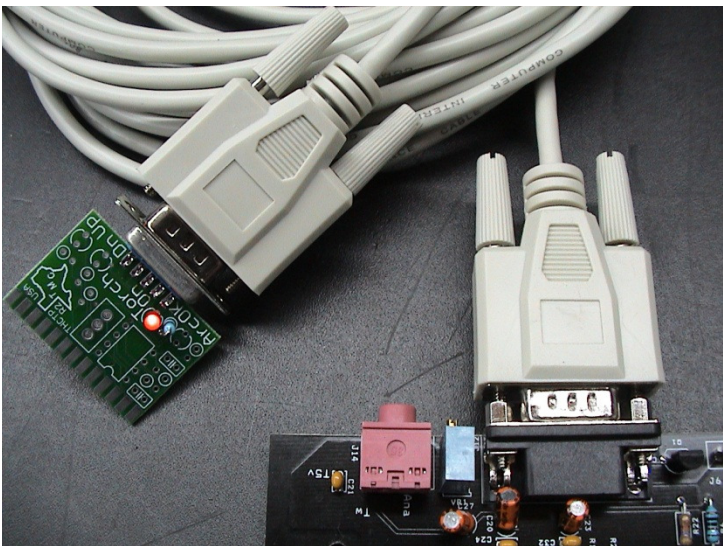


Figure 23 - Install DB9 Test Plug in 25ft Sensor Board Cable

Testing the Sensor Board

The THC Sensor Board is the interface to the myriad of different plasma cutters in the world. We are going to use the clip lead jumpers in the THC Test Set to simulate the plasma Tip Volts on the Sensor board and send it to the THC unit. We will also “wrap” the “Torch On” relay back into the “Arc OK” signal with the clip leads to simulate “Arc OK” to the THC. We will not be testing the HECS (Hall Effect Current Sensor) as it was tested and calibrated at the factory. However, we will be showing you how to connect the HECS to your plasma cutter should it not have the “Arc OK” signal available. For more detail or calibration information, see the User Guide.

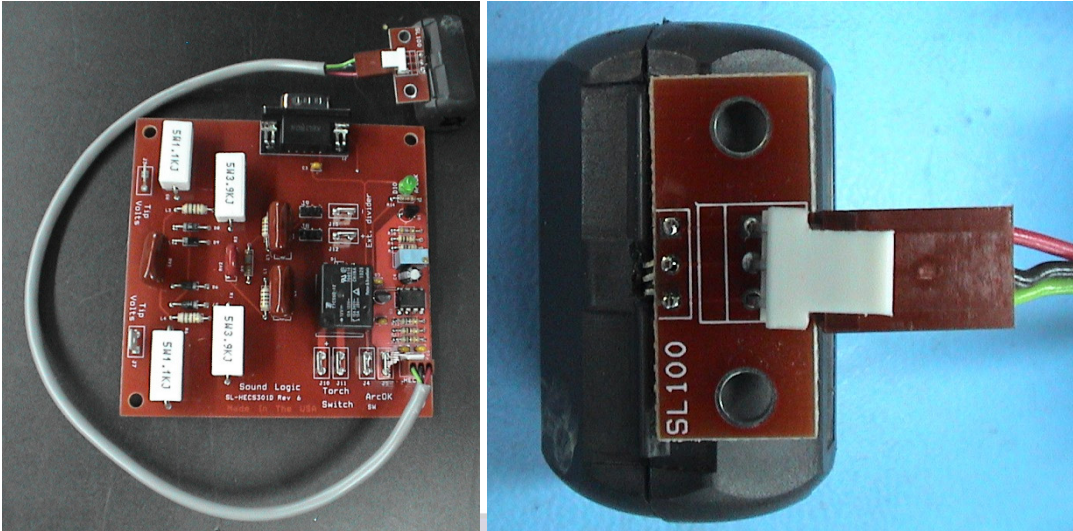


Figure 24 - Sensor Board with HECS

Sensor Board Locations

Below are locations of important points on the Sensor Board. For more detail on the function of these locations, see the User Guide.

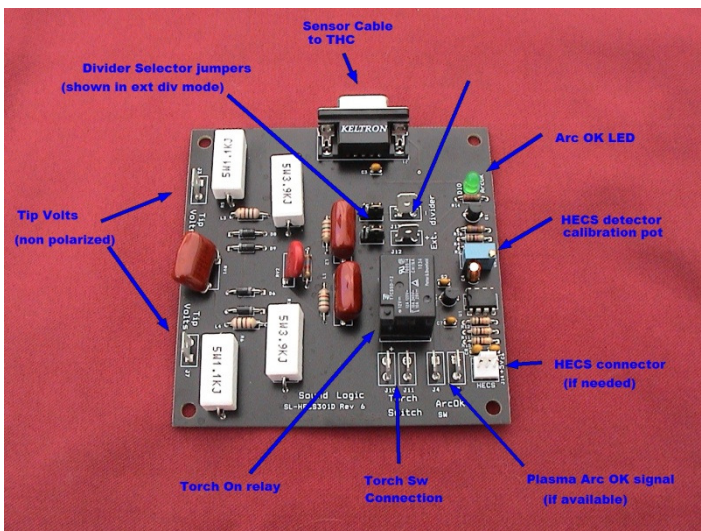


Figure 25 - Sensor Board Locations

Procedure to Test the Sensor Board

- First, locate the jumpers at J8 and J9 and temporarily move them from using the internal voltage divider circuit (left photo below) to using an external voltage divider circuit (right photo below).

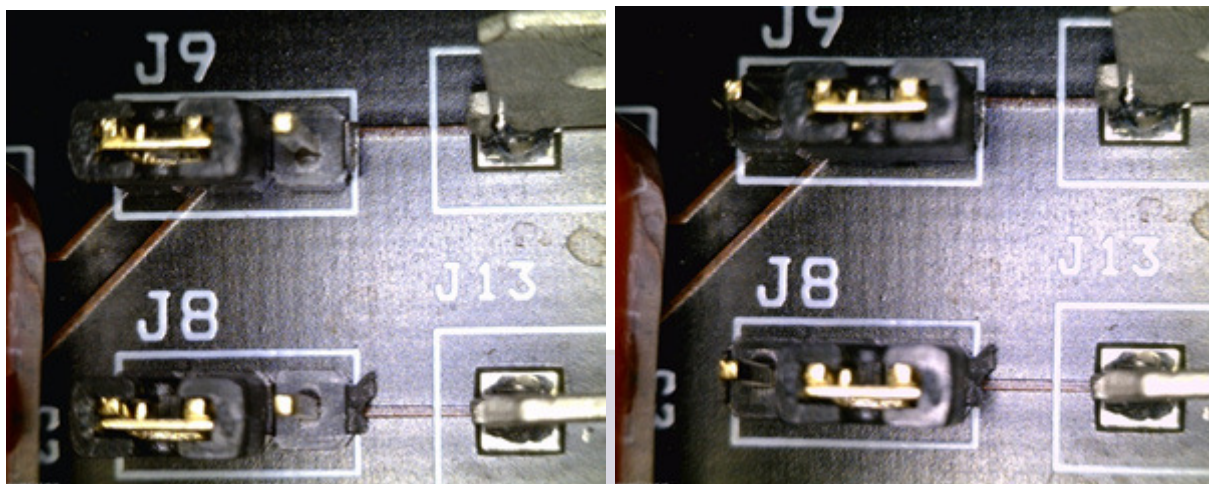


Figure 26 - Changing Voltage Divider Jumpers

- Next, using the Red clip lead, connect one end to the R14 resistor as in the photo below on the left. Connect the other end of the clip lead to the plus (+) Ext. divider connector at J12 as shown in the photo below on the right. Please pay close attention to the polarity.

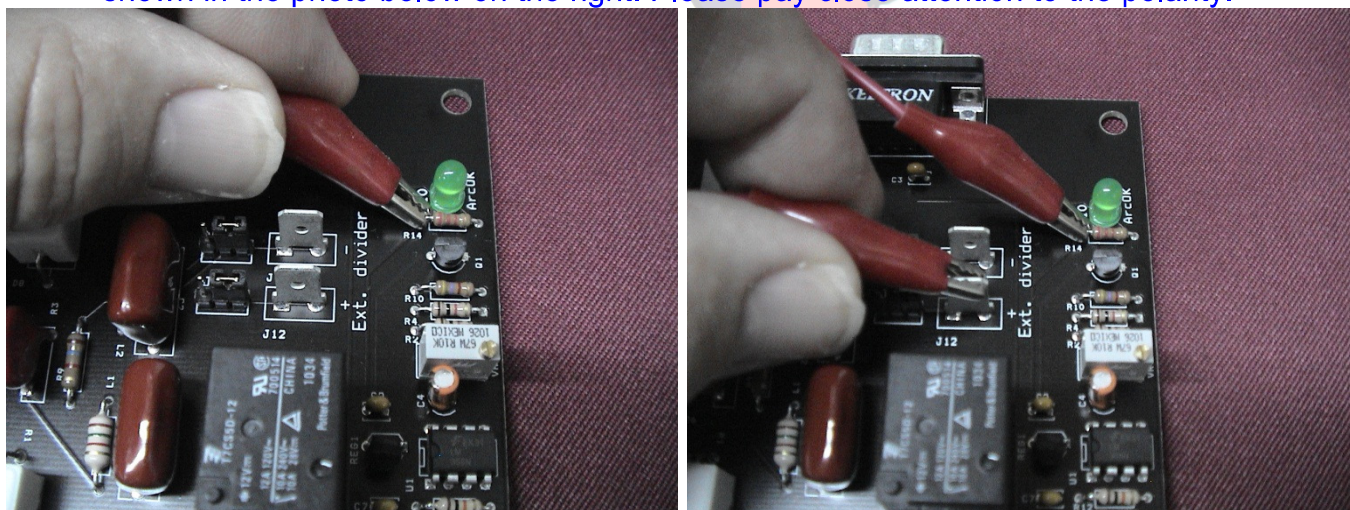


Figure 27 - Connecting Test Clip Lead to Plus (+) External Divider Terminal

Now connect the one end of the Black clip lead to the Ext. divider with the minus (-) on J13 as shown in the photo below on the left. Connect the other end for the clip lead to J5 of the “Arc OK SW” as shown in the photo below on the right. Please pay close attention to the polarity.

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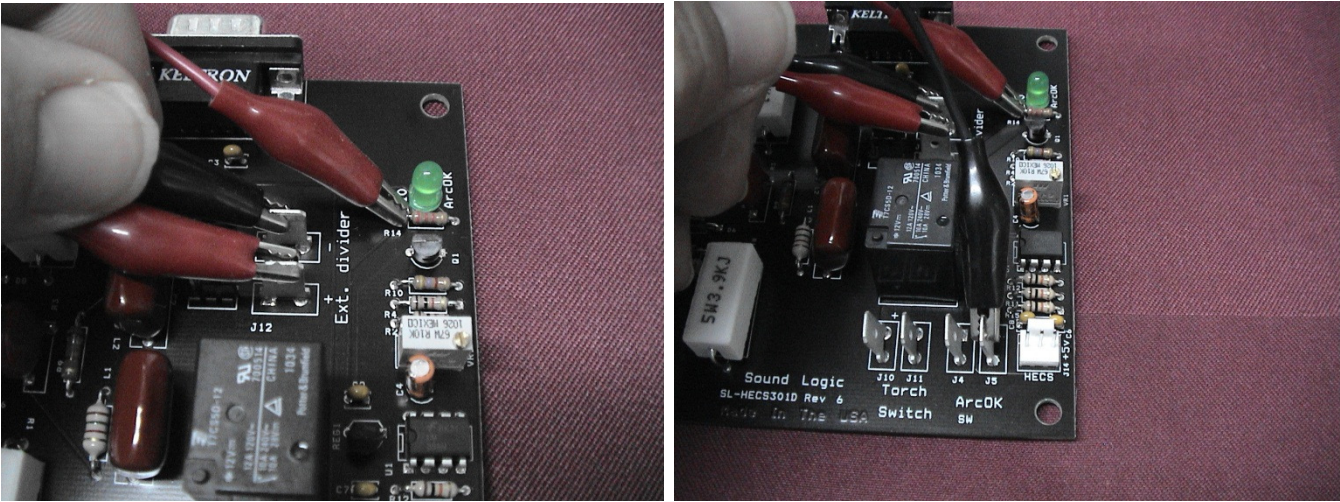


Figure 28 - Connecting Test Clip Lead to Minus (-) External Divider Terminal

Take the **Green** clip lead and connect J11 and J5. Take the **White** clip lead and connect J10 to J4. This will connect the Torch On signal to “wrap back” Arc OK and simulate the proper response from the plasma cutter to the Sensor Board.

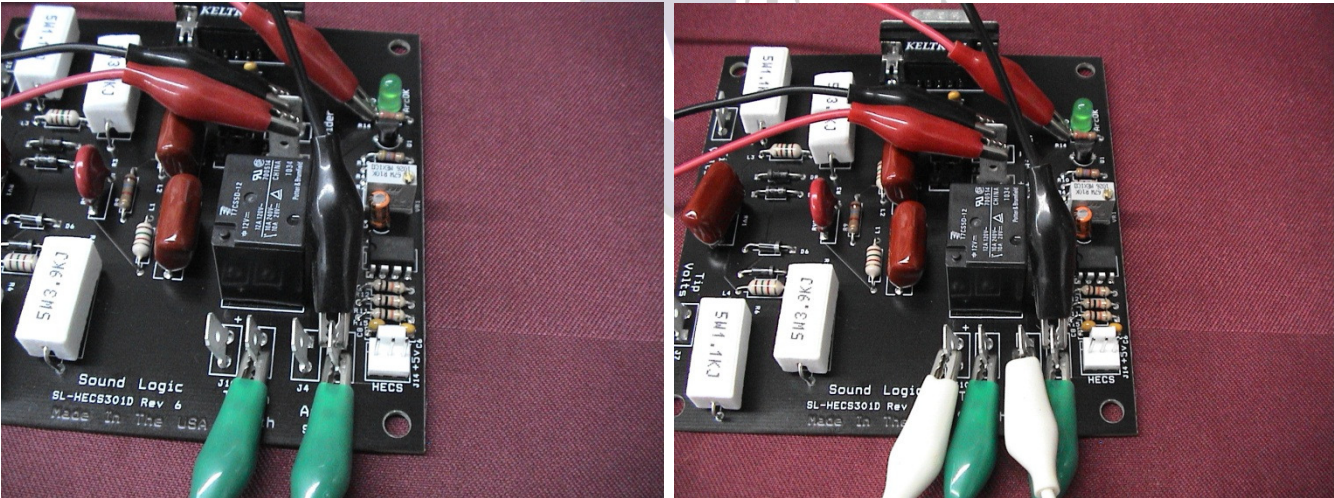


Figure 29 - Wrapping "Torch On" to "Arc Ok"

Now repeat the steps for the THC Bench Test and you will have completed an “end to end” testing of the entire THC system without connecting to the PC/Mach and the plasma cutter.

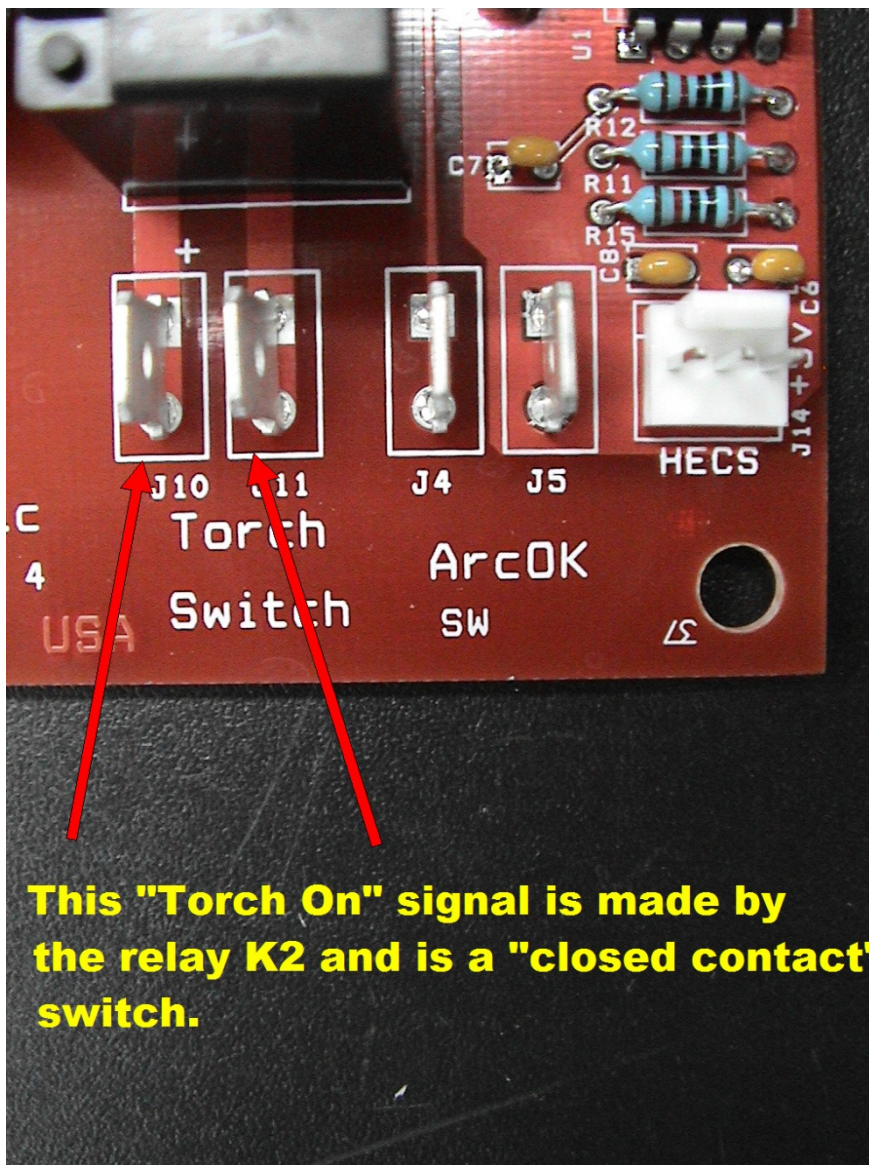
Don't forget to move the jumpers you moved in Figure 24 back to their default location like in the left photo of figure 24.

Connecting the Sensor Board to the Plasma Cutter.

This may take a little longer than the 10 minutes that was used to test the system as each plasma cutter is different and mounting skills vary from user to user. There will be no photos of the connection inside your plasma cutter model for connecting the “Torch On” and “Arc OK” signals because each plasma cutter is different and the connections are different even within the same manufacturer. We have designed the Sensor Board to be flexible and adaptable to just about every plasma cutter made.

There are 3 basic connections:

- The **Torch Switch** is the connection that sends the “Torch On” signal to the plasma cutter. This is a “closed contact” connection and is provided by a relay (K2) on the Sensor Board. Though your plasma cutter may provide the “Torch On” connection on a terminal strip or standard connector, you can also “trigger” the plasma cutter simply by connecting across the trigger switch on your hand torch of you plasma cutter.



This "Torch On" signal is made by the relay K2 and is a "closed contact" switch.

Figure 30 - Connecting Sensor Board to "Torch On" Signals

- The “**ArcOK**” or “**ArcGood**” or “**Ready to Move**”, signal is generated by the plasma cutter. This signal is usually a contact closure (like a relay switch) that uses closed contacts to produce the “ArcOK” condition. This ArcOK condition is valid once the plasma cutter transfers the arc from tip to electrode and the cutting current flows. When “ArcOK” is detected, the **Green** LED on the Sensor Board is illuminated.

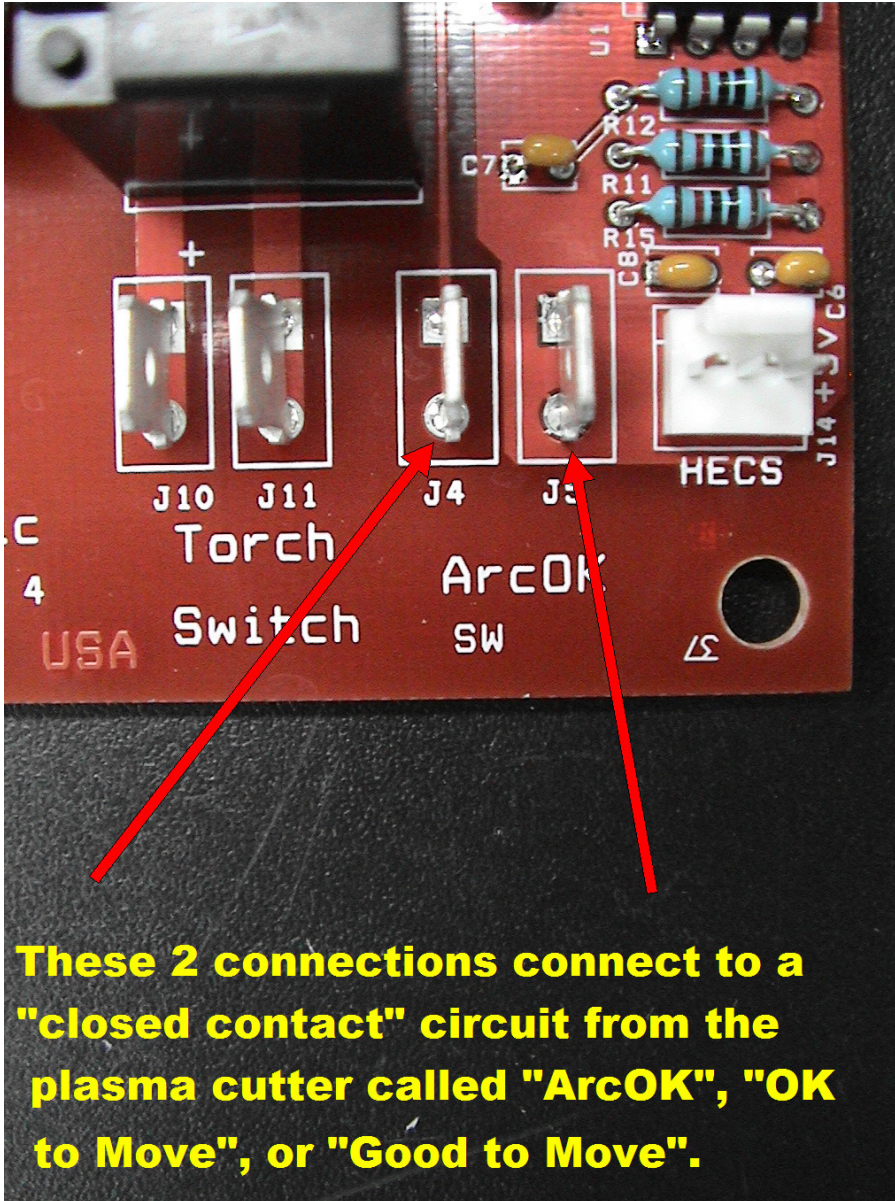


Figure 31 - Connecting Sensor Board to ArcOk Signal

- For those plasma cutters (usually older machines or ones that were designed for hand torches only) that do not generate an “ArcOK” signal, we can still detect when the arc has been established by reading when the cutting current is flowing. The Sensor Board has a Hall Effect Current Sensor circuit onboard. It is connected to a cable with a Hall Effect sensor and a ferrite core that clamps around the work cable (cable you connect to the material being cut) and this HECS detects when the cutting current starts to flow and generates the ArcOK signal on the

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Sensor Board. The HECS is easy to install as the 2 photos below will show. The HECS is magnetically triggered by the magnetic field that is generated by the current flowing through the work cable and induced in the ferrite core. If the HECS is installed and firing of the torch does not turn on the **Green** LED, you may need to reverse the HECS (just unclip it and turn it around) as it is polarized.

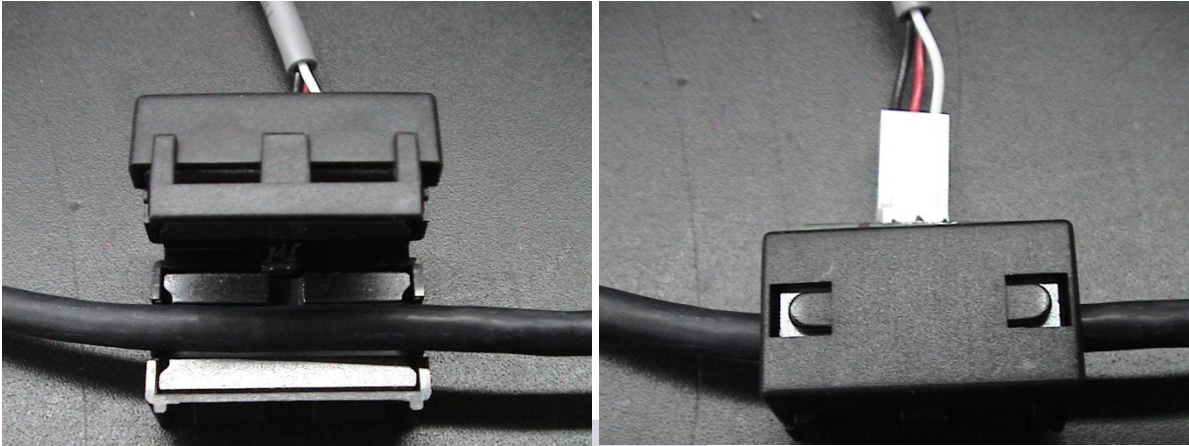


Figure 32 - Connecting HECS to Plasma Cutter

- The **Tip Voltage** is the high voltage (up to 300 or more volts) that is measured across the plasma arc. This voltage connection is retrieved by a connection inside the plasma cutter. One side of the Tip Voltage is the connection where the work cable connects to the electronics in the machine. The other is where the torch cable is connected in the machine. Care must be taken when working around these high voltages and always use a qualified technician to help if you feel you are not qualified to make these connections. You must use a #18 gauge or larger wire that is rated for 600 volts or more to make this connection. The Tip Voltage connections should go to J3 and J7 on the Sensor Board. They are non-polarized. More information using the newer (and safer) internal voltage divider Tip Voltage can be found in the User Guide.

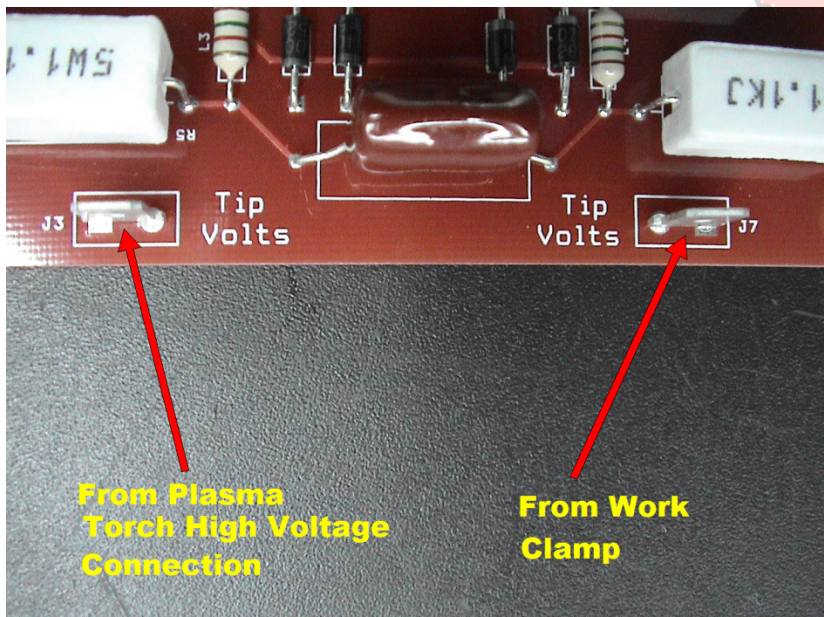


Figure 33 - Connecting the Plasma Cutter to Raw Tip Volts

Software Installation

Software installation for the THC is simple. It is assumed that you have already installed a licensed copy of Mach3 per the Mach3 installation instructions.

Download the THC301 (same for the THC300) install program from our website at:

<http://www.texasmicrocircuits.com/thcdoc.htm>

The install program is called “THC300 & THC301 Install” THC30x.exe or THC30x.zip

If you use the THC30x.exe, it will install in your Mach directory on the C: drive.

If you use the THC30x.zip, unzip it to your C: root and it will install in your Mach directory.

Mach Screen for the THC300 & THC301 Torch Height Controllers

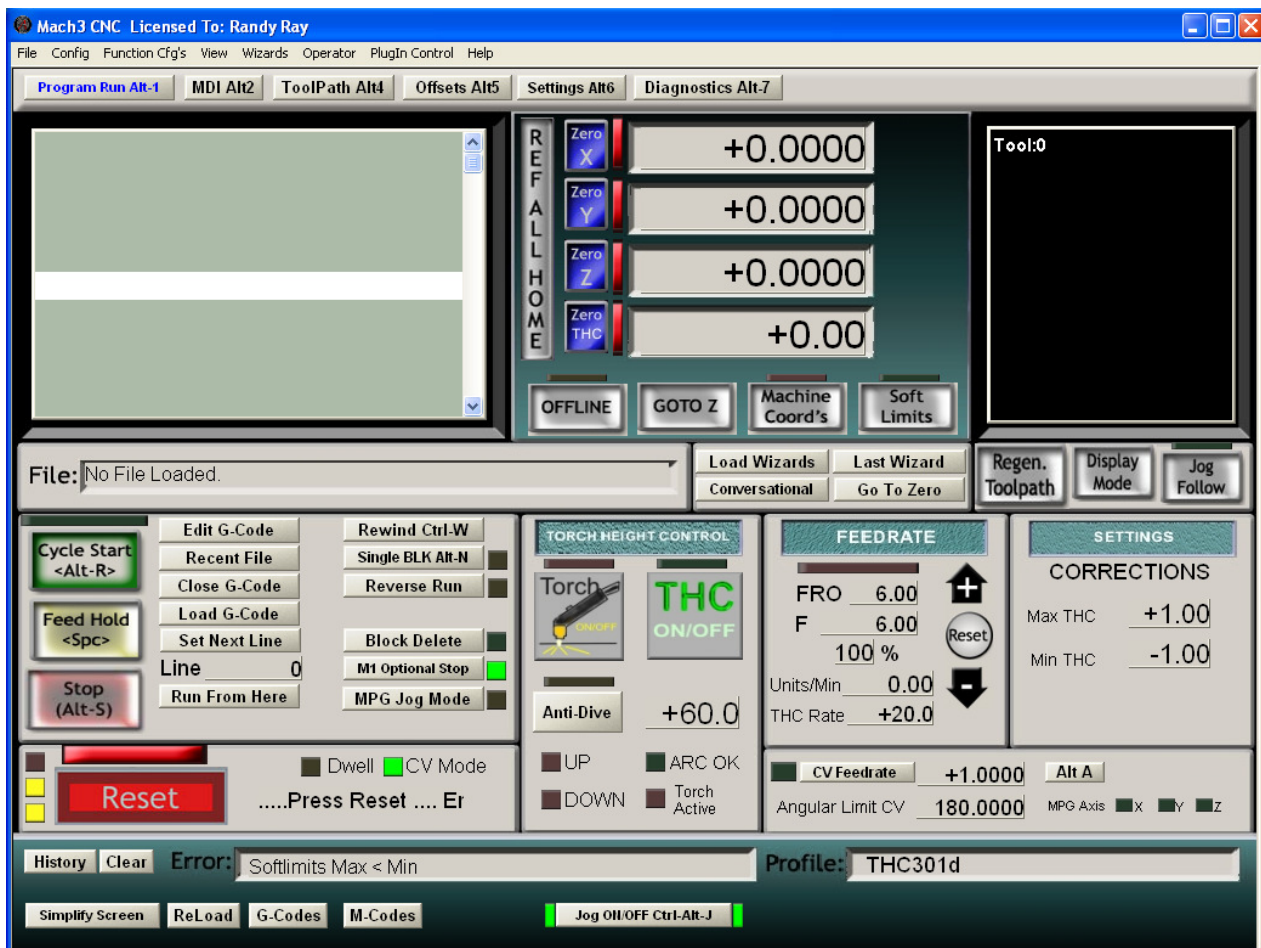


Figure 34 - Mach Screen for THC300 & THC301

Mach Screen THC Settings

Torch Height Control Area (Center Bottom)

Torch On/Off – This is both a “pushbutton” and indicator. It allows manually firing the torch by clicking on the Torch button and the red bar above the button will light. When Mach3 send the Torch On signal to the THC, the red bar also lights.

THC On/Off – This is the control activating the THC. When it is off, Mach3 ignores any commands from the THC. When it is on, Mach3 will depend on the THC for communication to maintain proper tip height of the torch. [Turn this button ON by clicking it.](#)

Anti-Dive – This button/indicator turns on/off the Mach3 **internal** Anti-Dive mechanism that keeps the THC from diving (moving the torch down when falsely detecting a higher Tip Voltage) when crossing a gap (previously cut line) or rounding corners. Though a few users have played with this, most don't think it provides any added benefit and rely entirely on the THC internal anti-dive mechanism.

[Recommendation is to leave this feature turned OFF.](#)

Anti-Dive DRO – This number is actually a **percentage** of the running velocity (XY combined) and is set in the Anti-dive DRO (to the right of the Anti-dive button). If activated and the average velocity of XY drops below the percentage set in the Anti-dive DRO, the Z axis will not move further down. Once again, the THC301d internal Anti-Dive firmware will work better than this Mach3 version.

[Recommendation is to leave this feature turned OFF.](#)

Up – Down – These indicators show when the Z axis is moving Up or Down.

ARC OK – This indicator shows when the THC reports to Mach3 that the Arc has transferred and Arc OK is valid.

Torch Active – I have no idea what this indicator does other than, possibly, the obvious.

Feed Rate Area (Right of Torch Height Control)

THC Rate – This number is a **percentage** of the set cutting feed rate and is used to throttle the Z axis movement and make it proportional to the feed rate. If the THC Rate is set to low, the Z axis will not keep up with sudden Arc Voltage changes such as when cutting corrugated material. If the THC Rate is set too high the Z axis becomes overly sensitive to Arc Voltage changes and the Z axis will oscillate up and down trying to the axis perfectly “in the zone”. [Set this value to 15 or 20 to start with.](#)

Settings Area (Right Bottom)

Max THC – This is the maximum limit the Z axis will be allowed to travel under THC control. [Set this value to +1 to start off.](#)

Min THC – This is the minimum limit the Z axis will be allowed to travel under THC control. [Set this value to -1 to start off.](#)

Operation

Torch Height Control Operation Sequence for THC301d

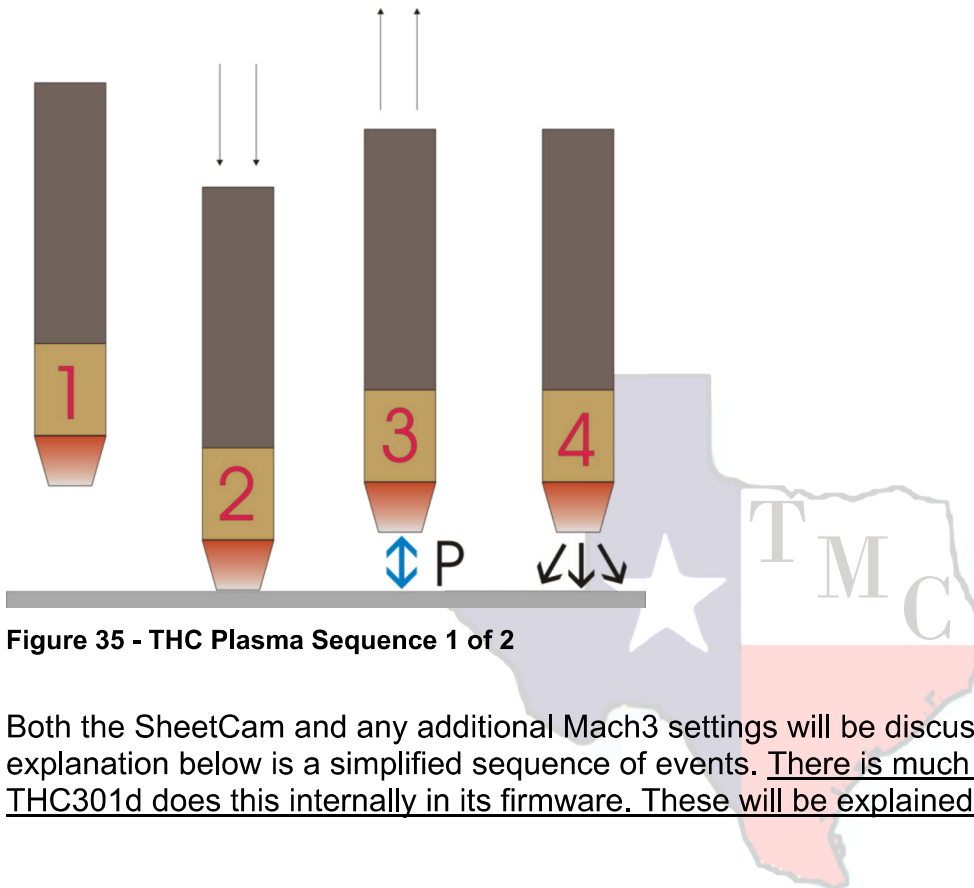


Figure 35 - THC Plasma Sequence 1 of 2

Both the SheetCam and any additional Mach3 settings will be discussed in the User Guide. The explanation below is a simplified sequence of events. There is much more to consider and the THC301d does this internally in its firmware. These will be explained in the User Guide.

Assuming Mach3 is set up properly for THC Plasma operation, the “Floating Head” touch-off is properly set up and functioning and a G-Code “cut file” has been properly generated from SheetCam, the sequence of events follows when plasma cutting:

1. The Plasma Torch is at some “retract” height above the work metal.
2. Pressing Start on Mach3 (with the THC On/Off button activated on the Mach3 screen under Torch Height Control) causes the torch to move to the requested X/Y position. The Z axis moves down and the “floating head switch” or ohmic sensor detects the work piece with the tip of the torch.
3. Mach3 then raises the torch to the Pierce Height (determined by settings in SheetCam) – P=Pierce Height
4. Mach3 then turns on the TORCH light under Torch Height Control and tells the THC to fire the torch (Torch On signal). The THC turns on the TORCH ON LED on the front panel then sends

the signal to the THC Sensor board where the relay activates and sends the “torch on” signal to the plasma cutter. Gas flow (shop air in most cases) begins.

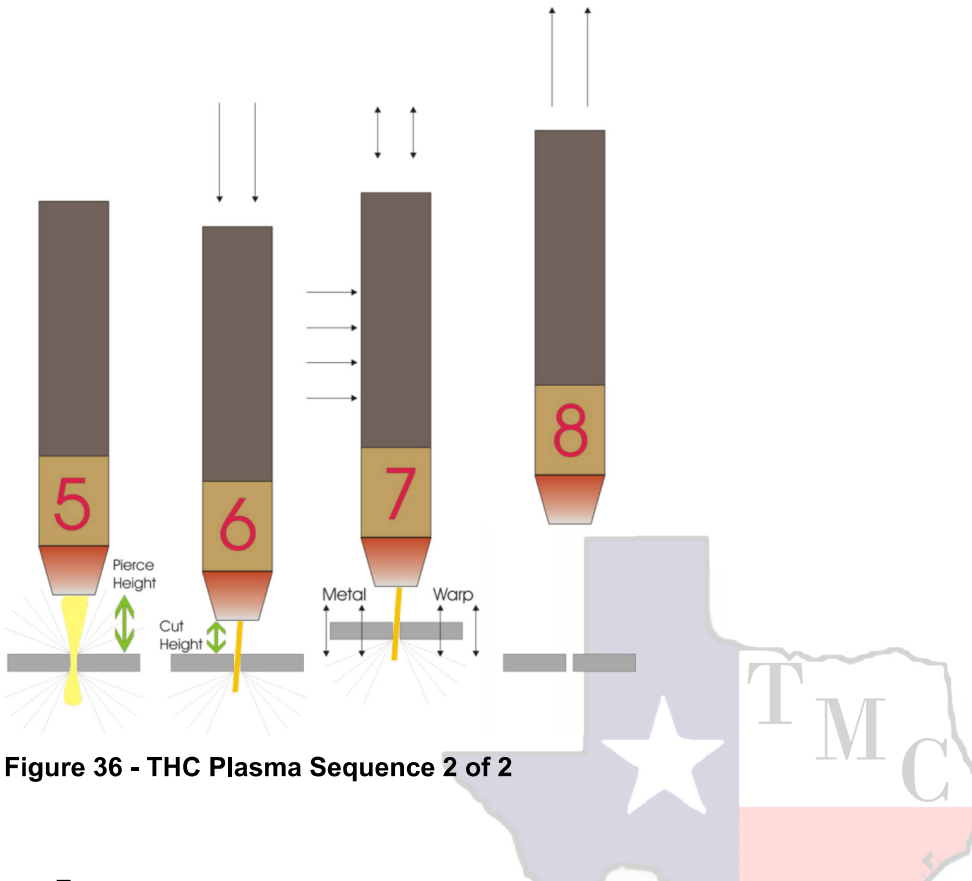


Figure 36 - THC Plasma Sequence 2 of 2

5. The plasma pilot arc initiates via the plasma machine. The pilot arc transfers to the cutting arc and cutting arc connects with the work piece. Arc OK signal is sent to the THC Sensor board via either an on-board “Arc OK” or “Arc Good” signal from the plasma cutter or the Hall Effect Current Sensor detects a current spike (when the transfer from pilot arc to cutting arc is executed) and the Green LED on the THC Sensor board lights. The “Arc OK” signal is then transferred to the THC where it turns on the ARC OK LED on the THC front panel and then sends it to Mach3 via the parallel port cable. Mach3 then turns on the ARC OK square “LED” on the screen in the Torch Height Control section. If there is a Pierce Time Delay that was set in SheetCam for this operation, then the Mach3 delays moving the torch until the delay has ended.
6. Now Mach3 working with the THC moves the torch down to the Cut Height as determined by the Arc Voltage. The Arc Voltage or Tip Voltage is sensed by the Voltage Divider Circuit on the THC Sensor board and sent to the THC. The THC compares it to the Set Voltage (Set Voltage is the number the user set in the THC by adjusting the TIP HEIGHT knob(s) on the THC front panel and reflected in the THC display) and tells Mach3 to adjust the Z axis to compensate. Mach3 now tells the torch to move and start cutting.
7. As the torch moves and cuts the work piece the Tip Voltage may change if the metal warps or is not perfectly flat. This necessitates movement of the Z axis to maintain the proper cut height. The THC tells Mach3 to move the Z axis up if the Set Voltage is higher than the Tip Voltage or

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down if the Set Voltage is higher than the Tip Voltage. If they are equal, then no Z axis motion is performed. These actions are reflected in both the UP and DOWN lights on the Mach3 screen and on the THC.

8. When the cut is finished, Mach3 stops X/Y movement, tells the THC to turn the torch off, whereas the THC loses Arc OK, and Mach3 raises the Z axis to the “retract” height. If there is more to cut, then Mach3 will move the torch to the next X/Y position and the process will start over.

That's it! If all has worked to this point, you are ready to start operations.

If there have been some inconsistencies in results, you will need to refer to the User Guide for more detailed information.

